Object Detection for Wildlife Conservation

Detecting Waterfowl in UAV Thermal Imagery

YOLOv8s Implementation - Computer Vision Assignment 1

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Dataset Overview

- 542 total images: 355 positive + 187 negative
- 8,975 bird annotations across positive images
- Average: 25.3 birds per image
- Image size: 512×640 pixels (7.5cm/pixel GSD)

Dataset Split

Dataset Split Plan:

Split	Positive	Negative	Total	Neg %
Train	266	141	407	34.6%
Val	53	28	81	34.6%
Test	36	18	54	33.3%

Pipeline: Preprocessing and Training

- Processing positive images (with birds) and their annotations
- Processing negative images (without birds) - important for **reducing false positives**
- Image quality checks and channel standardization
- Converting CSV annotations to **YOLO format**
- Splitting into **train/val/test** sets

💻 **Hardware**

- GPU: RTX 3050

Parameter	Your Value
Model	yolov8s.pt
Epochs	50
Batch Size	8
Image Size	640
Learning Rate	0.001
Optimizer	Adam
Patience	20
Workers	0
Device	Auto (GPU/CPU)

Evaluation Results (Validation Set)

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Model saved: runs\train\waterfowl_yolo\wei

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FINAL METRICS
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Best Epoch: 33/50
mAP@0.5:    0.9263
Precision:  0.8802
Recall:     0.8565
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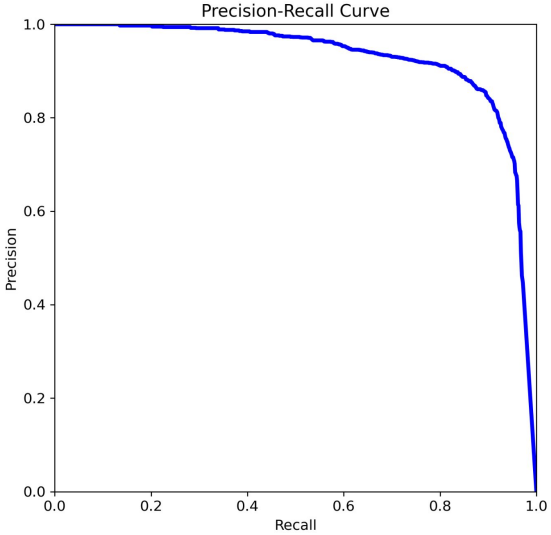

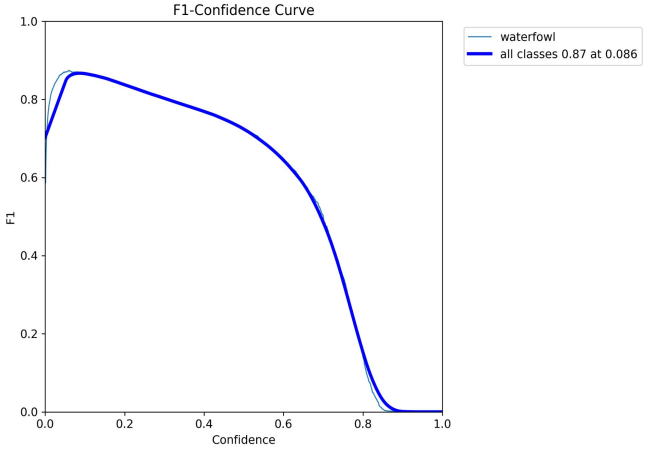

Error Analysis & Visualization (Test Set)

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EVALUATION METRICS
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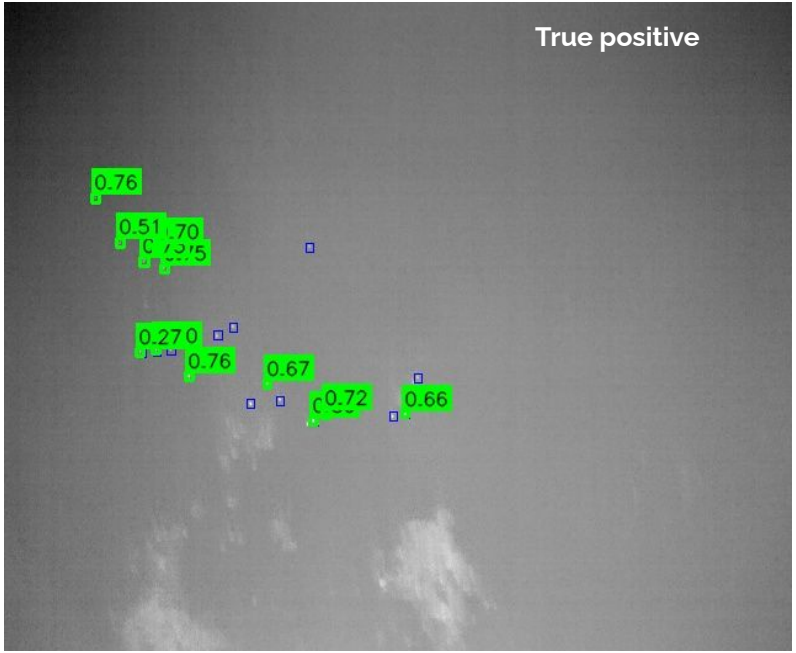
Detection Statistics:
  True Positives:  811
  False Positives: 51
  False Negatives: 277

Performance Metrics:
  Precision: 0.9408
  Recall:    0.7454
  F1-Score:  0.8318
```



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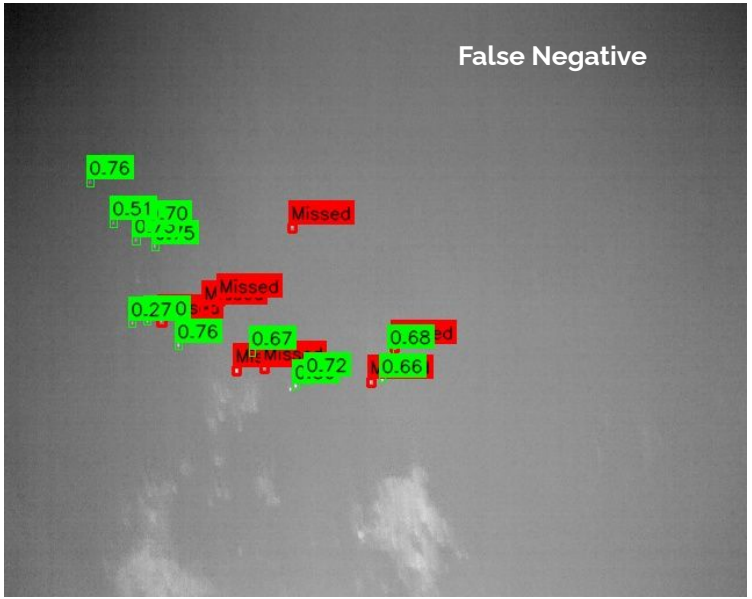

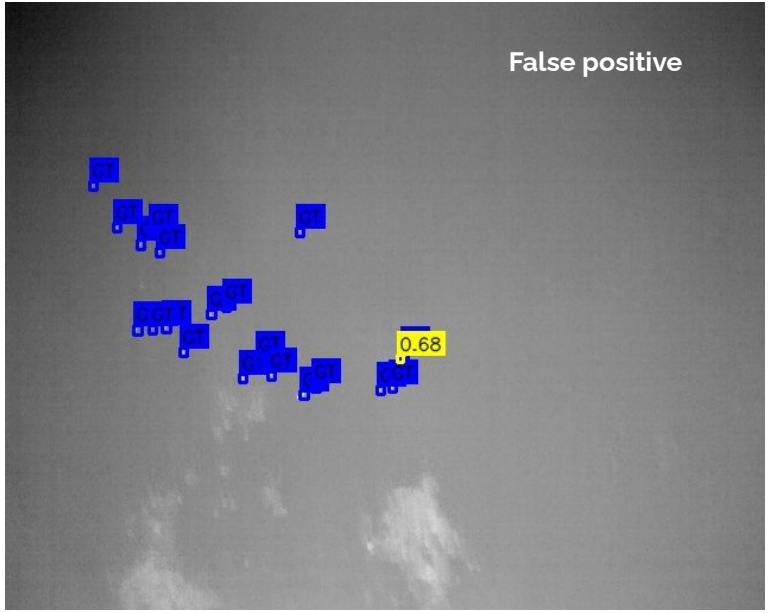

Thermal Imagery: Strengths & Challenges

Strengths

- Clear thermal signatures of warm-blooded waterfowl
- Effective in low-light conditions (day/night)
- Consistent across lighting conditions

Challenges

- Lower spatial resolution than RGB cameras
- Reduced texture and fine detail
- Can confuse warm objects (rocks, vegetation)
- Environmental temperature variations
- Difficulty with overlapping birds

Key Achievement

Successfully implemented YOLOv8s for waterfowl detection achieving 92.63% mAP on validation set with excellent precision.

Assignment 2

We consider the standard Triplet Loss defined for a triplet (a, p, n) (anchor, positive, negative) as

$$\mathcal{L} = \max\big(0,\ d(a,p) - d(a,n) + m\big),$$

where

- a is the anchor sample,
- p is a positive sample from the same class as a,
- n is a negative sample from a different class,
- $d(x, y)$ denotes the Euclidean distance between the embeddings of x and y,
- $m > 0$ is the margin parameter.

Intra-class and inter-class distances in the embedding space are defined as $\mathbb{E}[d(a,p)]$ and $\mathbb{E}[d(a,n)]$, respectively. We should show analytically how the choice of m affects the average inter-class distance $\mathbb{E}[d(a,n)]$ in the embedding space, and derive the condition under which $\mathbb{E}[\mathcal{L}] = 0$.

Task 1.1:

Effect of Margin m

Inside the max():

$$z = d(a,p) - d(a,n) + m.$$

- If $z \leq 0 \Rightarrow \mathcal{L} = 0$ (triplet is "satisfied", no gradient).
- If $z > 0$ (i.e. $d(a,n) < d(a,p) + m$) $\Rightarrow \mathcal{L} = z$ and
  - $d(a,p)$ is pushed to decrease (same-class points move closer),
  - $d(a,n)$ is pushed to increase (different-class points move apart).

Thus, a triplet produces a positive loss exactly when the negative sample is closer to the anchor than m beyond the positive sample. So training aims for each triplet

$$\mathcal{L} = 0 \quad \Longleftrightarrow \quad d(a,n) \geq d(a,p) + m.$$

Task 1.1:

Effect of Margin m

We introduced expected intra- and inter-class distances:

$$\mathbb{E}[d(a,p)] \quad \text{and} \quad \mathbb{E}[d(a,n)].$$

Also we received a condition for all triplets:

$$d(a,n) \geq d(a,p) + m,$$

then by taking expectations and using linearity we obtain:

$$\mathbb{E}[d(a,n)] \geq \mathbb{E}[d(a,p)] + m.$$

Conclusion:

- The margin m sets a lower bound on the difference between the average inter-class and intra-class distances.
- Larger m forces stronger separation between class clusters in the embedding space.

Task 1.1:

Condition for $\mathbb{E}[\mathcal{L}] = 0$

For any triplet

$$\mathcal{L}(a, p, n) = \max\big(0,\ d(a, p) - d(a, n) + m\big) \ \geq\ 0.$$

A non-negative random variable has zero expectation iff it is zero almost surely:

$$\mathbb{E}[\mathcal{L}] = 0 \iff \mathcal{L}(a, p, n) = 0 \text{ for (almost) all triplets.}$$

For a single triplet, zero loss is equivalent to a constraint on distances:

$$\mathcal{L}(a, p, n) = 0 \iff d(a, p) - d(a, n) + m \leq 0 \iff d(a, n) \geq d(a, p) + m.$$

Thus,

$$\boxed{\mathbb{E}[\mathcal{L}] = 0 \iff d(a, n) \geq d(a, p) + m \text{ for (almost) all triplets } (a, p, n)}.$$

Geometrically, all class clusters are separated in the embedding space by a margin of size m.

Task 1.2:

Too Small Margin

Assume the model reaches a state where, for almost all triplets, there exists $\Delta > 0$ such that

$$d(a, n) \geq d(a, p) + \Delta.$$

Choosing a too small margin from range $0 < m < \Delta$ leads to:

$$d(a, p) - d(a, n) + m \leq m - \Delta < 0,$$

hence

$$\mathcal{L}(a, p, n) = 0$$

for almost all triplets.

Implications:

- $\mathbb{E}[\mathcal{L}] \approx 0$: gradients vanish.
- Training effectively stops, although the separation can still be fragile.
- Too small m makes the model unstable and sensitive to small input changes.

Task 1.2:

Too Large Margin

Assume the embedding space is bounded:

$$0 \leq d(u, v) \leq D_{\max} \quad \text{for all } u, v.$$

Then for any triplet

$$d(a, n) - d(a, p) \leq D_{\max}.$$

Zero-loss condition:

$$\mathcal{L}(a, p, n) = 0 \iff d(a, n) - d(a, p) \geq m.$$

If we choose too large margin

$$m > D_{\max},$$

these two inequalities contradict each other.

Implications:

- No triplet can satisfy $\mathcal{L} = 0$.
- The desired zero-loss regime is unattainable: training cannot properly converge.

Task 2.1:

Embedding statistics

In our developed method for detection Out-of-Distribution we use the trained metric learning model as an embedding function:

$$f : x \mapsto z \in \mathbb{R}^d.$$

For each in-distribution (ID) class $c \in \mathcal{C}$ we define:

- Training embeddings: $z_i^{(c)} = f(x_i^{(c)})$.
- Class center:

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} z_i^{(c)}.$$

Each ID class (waterfowl species) forms a compact cluster around its prototype in the embedding space.

Task 2.1:

OOD Score and Decision Rule

For a new sample x:

1. Compute its embedding:
$$z = f(x).$$
2. Compute distances to all class prototypes:
$$d_c(z) = \|z - \mu_c\|, \quad c \in \mathcal{C}.$$
3. Define the OOD score as
$$\text{OODScore}(x) = \min_{c \in \mathcal{C}} d_c(z) = \min_{c \in \mathcal{C}} \|f(x) - \mu_c\|.$$

Decision rule:

$$\begin{cases} \text{OODScore}(x) \leq \tau & \Rightarrow x \text{ is ID}, \\ \text{OODScore}(x) > \tau & \Rightarrow x \text{ is OOD}, \end{cases}$$

where τ is calibrated on ID validation data.

Task 2.2:

Ideal Class Separation

We consider an idealized embedding space:

- $\mathcal{X}_{\text{ID}}$: set of ID samples (waterfowl images).
- $\mathcal{X}_{\text{OOD}}$: set of OOD samples (background, non-bird objects, other species).

Assumptions for ideal class separation:

1) There exists $r_{\text{ID}} > 0$ such that

$$\forall x \in \mathcal{X}_{\text{ID}} : \ \text{OODScore}(x) \leq r_{\text{ID}}.$$

2) There exists $r_{\text{OOD}} > r_{\text{ID}}$ such that

$$\forall x \in \mathcal{X}_{\text{OOD}} : \ \text{OODScore}(x) \geq r_{\text{OOD}}.$$

Geometric: ID clusters are compact around centers μ_c, OOD samples lie outside all clusters with a positive margin.

Task 2.2:

Justification of the OOD Score

From (1) and (2) assumptions:

$$\max_{x \in \mathcal{X}_{\text{ID}}} \text{OODScore}(x) \leq r_{\text{ID}}, \qquad \min_{x \in \mathcal{X}_{\text{OOD}}} \text{OODScore}(x) \geq r_{\text{OOD}}.$$

With $r_{\text{OOD}} > r_{\text{ID}}$:

$$\max_{x \in \mathcal{X}_{\text{ID}}} \text{OODScore}(x) < \min_{x \in \mathcal{X}_{\text{OOD}}} \text{OODScore}(x).$$

For any ID x_{ID} and any OOD x_{OOD}:

$$\text{OODScore}(x_{\text{ID}}) < \text{OODScore}(x_{\text{OOD}}).$$

Consequences:

- There exists a threshold τ that perfectly separates ID and OOD.
- This justifies using $\text{OODScore}(x)$ from Task 2.1 as a principled OOD detector under ideal separation.